The Bottleneck Universe: A Macro Investor's Atlas of Country-Level Export Concentration & Disruption-Driven Opportunities

TL;DR

- **The world's industrial system is held together by a surprisingly small number of single-country, single-facility, single-mineral, and single-strait dependencies.** Concentration is most extreme in (a) China's processing monopoly across 15+ critical minerals (rare earths 91% refining, gallium ~99%, magnesium ~95%, tungsten ~83%, graphite spherical 90%+), (b) Middle East energy chokepoints (Strait of Hormuz = 20% of global oil + ~30% of helium), (c) single-country agricultural inputs (Morocco 70% of phosphate reserves, Brazil 90% of niobium, Turkey 73% of boron, Russia/Belarus 40% of potash), and (d) single-firm processing nodes (CBMM 77% of niobium, ASML 100% of EUV, TSMC ~90% of leading-edge chips, VSMPO ~30% of aerospace titanium pre-war).
- **The 2026 sulfuric-acid cascade is the template.** Iran-war disruption to Middle Eastern sulfur (≈⅓ of seaborne sulfur) → tightened sulfuric acid → China's de-facto export ban from May 2026 → squeeze on Chilean/Congolese/Zambian copper, Indonesian nickel/HPAL, uranium ISL, and global phosphate fertilizer. The same Iran-Hormuz event simultaneously took ~30% of helium offline (Qatar Ras Laffan), threatening MRI uptime in 2-4 weeks for low-inventory hospitals and chip yields in Korea/Taiwan within months. Helium spot prices doubled in 14 days; antimony went from $1,400 to $38,000/t after China's August 2024 export controls (a 2,600% move). Quest Metals Each shock has a clear, narrow set of beneficiaries — the report below maps them.
- **For positioning, focus on the "second-best producer" trade plus the chemical-layer derivatives.** When China's primary mineral leverage is exercised, money flows to: Brazil (niobium, rare earth potential), Australia/Canada (lithium, uranium, REE — Lynas, Iluka, MP Materials, Cameco), Japan (titanium sponge, photoresists, fluorinated polyimides — Toray, Tokyo Ohka, JSR, Stella Chemifa, Osaka/Toho Titanium), Korea (memory, separators), Indonesia (nickel/HPAL cobalt), Morocco (phosphate/OCP), Turkey (boron/Eti Maden), and US/Canadian helium (Exxon, North American Helium). The non-obvious beneficiaries — sulfuric acid producers ex-China, fluorspar miners ex-China, antimony substitutes, ammonia capacity ex-Russia, chrome ferroalloy ex-South Africa, rail-transit operators ex-Suez — are where structural alpha sits.

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1. CRITICAL MINERALS & METALS BY COUNTRY DOMINANCE

China — the master node (USGS 2025; IEA 2025; White & Case 2024)

- **Rare earths**: ~69% of mined REE (2024 quota 270,000 t REO); China Briefing **~91% of global separation/refining**; International Energy Agency **~99% of processed dysprosium**, ~80% of NdPr; Benchmark Source **~98% of dysprosium oxide and terbium globally** (heavy REE almost monopoly via Jiangxi/Guangxi ionic-clay deposits Braumiller Law Group + 70% of medium/heavy feedstock from Myanmar's Kachin region — itself disrupted by Kachin Independence Army seizures in late 2024). Investing News Network Rare-earth magnets: China exported 58,000 t in 2024 — enough magnets for millions of EVs / thousands of wind turbines / military systems. International Energy Agency Heavy REE export controls since April 2025 (samarium, gadolinium, terbium, dysprosium, lutetium, scandium, yttrium); China Briefing ex-China dysprosium clearing $800–900/kg vs ~$240/kg domestic (2.5–3x premium). OilPrice.com HIDDEN DEPENDENCY: dysprosium and terbium are non-substitutable in NdFeB high-temp magnets used in jet engines, missile seekers, EV traction motors. **Zero commercial heavy-REE metallization exists outside China.** OilPrice.com
- **Gallium**: ~98.7% global. **Germanium**: ~60% global; export licensing since Aug 2023; 2024 exports –55% YoY, going mainly to Belgium (33%), Germany (32%), Russia (25%), Japan (6%). USGS
- **Magnesium**: ~95%. **Tungsten**: ~80–83% (US 0% domestic; Almonty's South Korean Sangdong restart key). **Antimony**: ~48% mining, dominant refining (price spike from $1,400→$38,000/t between Jul–Sept 2024 → $51,500–60,000/t by mid-2025; 2,600% rally; Quest Metals US gets 63% of imports from China). Visual Capitalist Center for Strategic and Internation **Bismuth, indium**: dominant. **Fluorspar**: dominant; CRM-listed by US in 2025. **Graphite**: 70% mined natural, Visual Capitalist 78% all forms, Investing News Network **~90%+ spherical/anode-grade**, Benchmarkminerals 86% global anode material. Visual Capitalist **Tellurium, molybdenum, manganese alloys**: significant.
- **Refining dominance** (independent of mining): cobalt refining ~70% of global, lithium chemicals ~65%, NCM precursor 91%, LCO cathode 84%, battery-grade cobalt sulphate 85%. Springer

DRC (Democratic Republic of Congo)

- **Cobalt**: 72–76% of global mine output (2024–25); 2024 production ~244 kt; Mining Technology CMOC's KFM (Kisanfu) + TFM (Tenke Fungurume) Lexology Mining Technology > 50% of country output. Quota system from Q4 2025 caps annual exports at ~96,600 t (≈half 2024 levels). Investing News Network Pricing rallied >$47,000/t Oct 2025. Investing News Network

- **Copper**: ~5% global mining; growing fast; cobalt is the by-product driver.
- **Coltan/tantalum**: dominant African producer (key for capacitors).

Indonesia

- **Nickel**: ~50% of global (2024) — export ban on raw ore drove HPAL/MHP buildout funded by Chinese capital. Lexology
- **Cobalt**: 31 kt (2024), ~10% global — emerging #2; Investing News Network 89% from MHP; Cobalt Institute 95% of supply growth to 2030 will come from MHP. Cobalt Institute
- **Tin**: ~20% global. **Bauxite ban** since 2023 redirected China dependence to Guinea.

Chile / Peru

- **Chile**: 23% global copper (5.3 Mt 2025); world's largest copper reserves (180 Mt). Investing News Network **Lithium**: ~65% of global lithium-carbonate export value (2024); ANDAMAN PARTNERS 71% of lithium exports go to China (also 74% of copper exports). ANDAMAN PARTNERS
- **Peru**: 3rd-largest copper (2.7 Mt, 2025); Investing News Network 9.1% of proven reserves. The National Interest Cerro Verde + Antamina ≈ ⅓ of country output. Investing News Network **HIDDEN DEPENDENCY**: ~20% of Chilean copper output uses sulfuric acid leach — directly exposed to the 2026 China sulfuric-acid export ban (Chile imports >1 mt/yr from China). MINING.COM

Australia

- **Lithium**: ~50% of global mined supply (Greenbushes, Pilgangoora). **Iron ore**: 54.6% of global iron-ore export value 2024 Worlds Top Exports (Pilbara). **Bauxite**: ~22.7% of global production 2025. Mining Weekly **Uranium**: ~10%. **REE** (Lynas, Iluka): only meaningful non-China light REE separator (Mt Weld + Kalgoorlie + Eneabba refinery).

Russia

- **Palladium**: 37–44% global The Week (Nornickel) — historic spike to $3,177/oz in March 2022. Phoenix Refining **Platinum**: 10–15%. **Nickel**: 5–10%. **Titanium**: VSMPO-Avisma was ~30% of global aerospace titanium; Substack pre-2022 supplied ~60% of Airbus and ~80% of Boeing titanium. Quest Metals Sponge output cut from 32,000 t to ~17,000 t post-war. **Aerospace-grade neon**: previously 50%+ produced in Russia, purified in Ukraine. Substack **Uranium enrichment**: ~40–46% of global SWU capacity (Tenex/Rosatom). **Crude/gas/diesel/coking coal**: see §3. **Cobalt**: 4%, declining.

South Africa

- **Platinum**: 71.5% global mined (2024); Mining Technology Bushveld Complex hosts ~80% of world PGM-bearing ore. EBSCO **Rhodium**: ~80%+ (USGS-ranked #2 critical mineral globally for 2025). **Chromium**: ~42.7% mining, Mining Technology holds **>72% of world chrome reserves**; Fastmarkets supplies 81% of China's chrome ore imports. Fastmarkets **Manganese**: ~40% (highest-grade); 80% of resources, 37% of viable reserves. Fastmarkets **Vanadium**: world leader. **Coal**: meaningful seaborne thermal exporter.

Guinea

- **Bauxite**: ~28–30% of global production; **supplies 69–70% of China's bauxite imports** Abrasivestocks (2024 158.7 mt, of which ~110 mt from Guinea). Q1 2025 exports surged 39% to 48.6 mt; Qiraat Africa annualized 199 mt for 2025. Mining Technology Qiraat Africa SMB (Chinese-controlled), Discovery Alert Chalco/Chinalco, AGB2A/SDM dominate. Mining Technology HIDDEN DEPENDENCY: with Indonesia's ore ban, Chinese aluminum is now structurally hostage to Conakry military government's licensing decisions.

Ukraine

- **Neon (semiconductor-grade)**: 70–90% of global pre-2022 (Iceblick, Ingas, Cryoin); Substack 2014 Crimea crisis → 7–10x neon prices. Substack **Titanium ore (ilmenite/rutile)**: significant. **Manganese, iron ore, sunflower oil, wheat, corn**: see §4.

Brazil

- **Niobium**: ~88–90% of global production USGS (USGS 2024); **CBMM alone holds ~77.4% global market share** Intelmarketresearch out of Araxá (Minas Gerais), with 70%+ of world reserves. Intelmarketresearch Capacity 150,000 t/yr ferroniobium. Marketreportsworld Niobium ranked #10 on USGS 2025 high-risk critical mineral list. USGS **Iron ore**: 22% global export value (Vale). **Bauxite, graphite, REE potential**.

Kazakhstan

- **Uranium**: 39–43% global World Nuclear Association (Kazatomprom; ISL via Inkai, Tortkuduk, Myunkum). World's largest by far. Wikipedia **Chromium ore**: top-3 producer. **Titanium minerals**: significant. EU has positioned Kazatomprom as Russian Rosatom replacement; Wikipedia Trans-Caspian route avoids Russia. World Nuclear Association

Africa Other

- **Mozambique**: large flake graphite (Balama – Syrah Resources), Gotrays Graphite titanium minerals, ruby (>80% global from Montepuez).
- **Madagascar**: ~75–80% of natural vanilla globally, fast-growing flake graphite (Molo –

NextSource), Fastmarkets cobalt (small), nickel.

- **Namibia**: 12% global uranium; World Nuclear Association Husab + Rössing.
- **Niger**: ~5–6% uranium; Orano-controlled Arlit/Somaïr/Imouraren Nuclear-free — destabilized by 2023 coup.
- **Cote d'Ivoire/Ghana**: 60%+ global cocoa (Cote d'Ivoire ~30% alone). World Population Review
- **Zambia**: 4.2% global copper; Mining Technology rising.

Other notable

- **Morocco**: ~70% of world phosphate-rock reserves (50 bn t); IntelliNews OCP holds 31% global phosphate-product market share, Wikipedia 54% of phosphoric-acid trade, 34% of phosphate rock trade. USGS At current rates, reserves last 1,300+ years. USAPP Ranked #1 to benefit from any disruption to Russian fertilizer.
- **Turkey**: **73% of world boron reserves** Enerji (Eti Maden, ~60% of global supply); Türkiye Today state-owned monopoly; >$1 bn/yr exports Türkiye Today primarily to China (36%), US, India. DP World Critical for fiberglass, ceramics, agriculture, Nd-Fe-B magnets.
- **Mongolia**: significant coking coal to China; Oyu Tolgoi copper; potential REE.
- **Finland**: nickel (Talvivaara/Terrafame), chromium (Outokumpu Kemi, only EU chrome mine).
- **New Caledonia**: ~5–6% nickel; political instability.
- **Norway**: 27% of EU pipeline gas; titanium minerals.
- **Argentina**: lithium (projected to overtake Chile as #2 by 2028, 21% CAGR). Mining Technology
- **Vietnam, Thailand**: emerging REE producers (combined ~7% of 2024 production).

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2. OVERLOOKED INDUSTRIAL INPUTS & SPECIALTY CHEMICALS

Sulfuric acid (THE template chain — already triggered)

- China = >40% of global sulfuric acid output. Akin Gump Strauss Hauer & Feld LL Halted exports effective May 2026 in response to Iran-war disruption to Middle East sulfur

MINING.COM (≈⅓ global, ~½ seaborne). Pre-shock China exported 1.3 mt Jan–Apr 2025; quota cut to 700,000 t for same period 2026; March 2026 shipments to Chile = 0 (vs 151,268 t in March 2025). Akin Gump Strauss Hauer & Feld LL FOB ME sulfur jumped from ~$495 to $695–700/t. **Cascading hits**: Chilean copper SX-EW (~20% of Chile's output), DRC/Zambia copper-cobalt, Indonesian HPAL nickel/cobalt, uranium ISL leaching, S&P Global phosphate fertilizer (China protecting domestic spring planting). Akin Gump Strauss Hauer & Feld LL **HIDDEN DEPENDENCY**: there is no near-term ex-China replacement at 100s-of-millions-of-tons scale. S&P Global

Hydrofluoric acid / fluorspar

- **High-purity HF (semiconductor)**: Stella Chemifa (Japan) ~63%, Daikin ~21%, Morita ~9% — Japanese trio = >93% global. Globalmarketmonitor South Korea totally exposed (2019 incident). **Crude HF/fluorspar**: China dominant (mining + processing) — fluorspar on USGS 2025 critical list.

Helium (Iran 2026 shock — active)

- Pre-shock: US ~40–43% (largest), Discovery Alert **Qatar ~30% (Ras Laffan)**, Fortune Algeria ~7%, Russia (sanctioned). Iran missile/drone strikes on Ras Laffan in Mar 2026 shut LNG + helium production; Discovery Alert QatarEnergy declared force majeure, Foreign Policy cut annual helium exports 14%; Fortune Strait of Hormuz closure stranded ~200 specialized cryogenic containers Discovery Alert ($1m each), All-Weather Media causing boil-off losses of 15–20% on rerouted cargoes. Discovery Alert South Korea sources 64.7% of helium from Qatar Al Jazeera (semis); All-Weather Media Taiwan, Japan, Singapore similarly exposed. MRI uptime begins to fail in 2–4 weeks for low-inventory hospitals. Discovery Alert Spot prices doubled in 14 days. J2 Sourcing US Federal Helium Reserve **finalized privatization in 2024**, Foreign Policy removing the historical buffer. J2 Sourcing **Beneficiaries**: ExxonMobil (LaBarge WY), North American Helium, Helix Exploration, Blue Star Helium, Al Jazeera Air Liquide German salt-cavern storage.

Neon, krypton, xenon (semiconductor noble gases)

- 70–90% of global semi-grade neon historically from Ukraine; Substack produced as byproduct of Russian/Ukrainian steel mill air-separation units. Substack **Neon = ~45% of demand for ArF/KrF excimer laser lithography**. ASN Plus Crimea 2014 → 7–10x prices; Substack 2022 invasion brought further dislocation. Ukrainian players: Iceblick (Odesa), Ingas (Mariupol — destroyed/damaged), Cryoin (Odesa). Diversification underway in China/Korea but multi-year build.

Specialty chip chemicals (Japan)

- **Photoresists**: Japan ~90% global; Bloomberg <5 makers (JSR, Tokyo Ohka Kogyo, Shin-

Etsu, Sumitomo, Fujifilm). EUV resists almost wholly Japanese. China rumor of November 2025 Canon/Nikon/Mitsubishi Chemical photoresist halt Asia Times sent Chinese photoresist stocks +14–46% in days. Asia Times

- **Fluorinated polyimides**: Japan ~90% (Sumitomo, Shin-Etsu, JSR, Globalmarketmonitor Daikin) — flexible OLED/Samsung-LG dependent. 93.7% of Korean imports from Japan. USITC

TDI / MDI (polyurethane intermediates)

- Highly concentrated: BASF (Germany), Covestro (Germany), Wanhua (China), Huntsman (US), Dow. Chinese capacity ~45% of global MDI; any China nameplate disruption (or environmental shutdown — common) tightens insulation/automotive/footwear chains.

Catalysts & PGMs

- Auto-catalyst PGMs: South Africa + Russia >75% global. Refining catalyst PGMs (Rh, Ru, Ir): South Africa overwhelmingly dominant. **Rhodium ranked #2 on 2025 USGS critical list** — South Africa is single point of failure.

Tungsten carbide tooling

- China >80% mining + processing. Rare Earth Exchanges Exports cut 13.75% YoY Jan–Sept 2025; Discovery Alert new MOFCOM export licensing 2026–2027. Rare Earth Exchanges US has 0% domestic tungsten production; Almonty's Sangdong (Korea) restart 2025 with 45% offtake committed to US is the largest non-China play. Center for Strategic and Internation

Carbon fiber

- **Toray (Japan) ~30% global** + Mitsubishi Chemical, Teijin = Japan ≈50%; Hexcel (US, Stamford); SGL (Germany); Zoltek (Toray-owned). Aerospace prepreg almost wholly Japanese-American. Boeing/Airbus dependency on Hexcel HM63 and Toray T800/T1100.

Aramid (Kevlar/Twaron/Heracron)

- Teijin 17.6% global, top-5 (Teijin, Kolon, Yantai Tayho, Hyosung, Toray) = 37.1%. Global Market Insights DuPont still significant in para-aramid. Body armor/optical fiber strength members concentrated.

Specialty steels & alloys

- **Aerospace nickel superalloys**: Allegheny Technologies, Howmet, Quest Metals Haynes International, Special Metals (US), VDM (Germany). **Stainless precision tube**: Sandvik (Sweden), Schoeller-Bleckmann (Austria) — dominant in oil&gas downhole.

Magnetic materials

- **NdFeB sintered magnets**: China ~92% globally; Japan (Shin-Etsu, Hitachi/Proterial) the only meaningful non-China supplier; US (MP Materials, REalloys) just spinning up; pricing premium ex-China 2.5–3x.

Refractories & specialty ceramics

- **Magnesia refractories**: China ~70% (Liaoning fused magnesia). **Silicon carbide abrasives**: China dominant. **Yttria-stabilized zirconia**: Saint-Gobain, Tosoh (Japan) significant.

Industrial diamonds & superhard materials

- Russia (Alrosa) ~30% natural diamond. Synthetic industrial diamond: China ~80%+ (Element Six is Western standout). Listed under China's "superhard materials" export controls Globaltradealert Aug 2024. Center for Strategic and Internation

Specialty glass

- **LCD glass substrate**: Corning (US), AGC (Japan), Nippon Electric Glass, Tunghsu (China) — oligopoly of 4. **Optical fiber preforms**: Corning, Prysmian, Furukawa (Sumitomo), Fujikura — critical for AI/data center buildout.

Precision bearings

- **Aerospace/precision**: SKF (Sweden), Schaeffler (Germany), NSK/NTN/JTEKT (Japan), Timken (US). **Wind-turbine main bearings**: very few qualified suppliers; SKF and Schaeffler dominate.

Fasteners & manufacturing inputs (the truly obscure)

- **Aerospace fasteners**: Howmet/Lisi/Precision Castparts — single-source on many Boeing/Airbus airframe parts.
- **Tire bead wire / steel cord**: Bekaert (Belgium) ~25–30% global.
- **Lead acid battery separators**: Daramic (US), Entek concentrated.
- **Lithium battery separators**: Asahi Kasei (Japan), Toray, SK ie technology, Senior (China). Wet-process separator highly concentrated.
- **Cigarette filter tow / vape coil wire**: Kanthal (Sweden) dominant.
- **Specialty zircon (nuclear grade)**: Iluka (Australia), Tronox + China refiners.

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3. OIL, GAS, AND ENERGY COMMODITIES BY EXPORTER

Crude oil (2024)

- Saudi Arabia $191 bn (15.2%), Worlds Top Exports Russia, USA, UAE, Canada — top 5 = 52% of global $1.26 tn trade. Worlds Top Exports OPEC = 50.8% of value. Worlds Top Exports Saudi 10.88 mb/d production 2024; Investing News Network Iraq 4.4 mb/d; UAE 2.9; Kuwait 2.5; U.S. Energy Information Administrat Iran ~3.0 (sanctioned). Russia 9.2 mb/d, U.S. Energy Information Administrat redirected to China/India.
- Strait of Hormuz: 20 mb/d in 2024 = ~20% of global liquids; U.S. Energy Information Administrat 84% destined for Asia. Wikipedia Saudi 5.5 mb/d through it. U.S. Energy Information Administrat Saudi East-West pipeline (5 mb/d to Yanbu) + UAE Fujairah pipeline (1.8 mb/d) U.S. Energy Information Administrat = ~2.6 mb/d realistic Hormuz bypass. U.S. Energy Information Administrat

LNG (2024)

- US 88.4 mt (#1), Australia 81 mt, Qatar 77.2 mt SAFETY4SEA — top 3 = 60% of global. Russia 33.5 mt; Malaysia 27.7 mt. SAFETY4SEA By 2030, US could be ⅓ of global LNG supply. IEA Qatar's entire helium + half its LNG flows through Hormuz.

Pipeline gas

- US 206 bcm; Russia 152 bcm (still); Norway 126 bcm World ranking sites (29% of EU imports H1 2025); IEEFA Qatar 126 bcm; Australia 106 bcm; Algeria 49 bcm. World ranking sites EU 2025 mix: Norway 29%, US 27%, Russia 13%, Algeria 12%. IEEFA Russian LNG ban: short-term contracts within 6 months of EU sanction entry, long-term from Jan 1, 2027. IEEFA

Refined product chokepoints

- **Diesel**: Russia previously huge to Europe (sanctioned/redirected); India (Reliance Jamnagar, Nayara) and Middle East (Jubail, Ruwais) the key swing exporters. Houthi disruption to Bab al-Mandeb has significantly tightened East-of-Suez diesel arb.
- **Jet fuel**: South Korea, India, Middle East are the marginal suppliers to Europe.
- **Refining clusters at risk**: Houston ship channel, Antwerp-Rotterdam-Amsterdam (ARA), Singapore (Jurong), Ras Tanura/Jubail/Ruwais.

Coking coal

- Australia ~55% of seaborne coking coal; Russia ~15%; US, Canada, Mozambique. Critical for Indian/Korean/Japanese steelmaking.

Thermal coal flows

- Indonesia (largest seaborne exporter), Australia, Russia, South Africa, Colombia.

Uranium fuel cycle

- **Mining**: Kazakhstan 39%, Canada 24%, Namibia 12%, World Nuclear Association Australia 10%, Niger 6%. **Conversion**: Russia (Rosatom), Cameco (Canada), Orano (France), CNNC (China). **Enrichment**: Russia ~40–46% global SWU. **Fuel fabrication**: Westinghouse, Framatome, TVEL, GNF.

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4. AGRICULTURAL AND FOOD COMMODITIES

- **Russia/Ukraine combined (pre-war)**: ~34% global wheat, 17% corn, 73% sunflower oil, 27% barley. PubMed Central
- **Russia**: largest nitrogen exporter (16.5% global N, American Farm Bureau Federation 15–16% global urea, 12% phosphate, 16.5% potash). American Farm Bureau Federation **Russia + Belarus = ~40% global potash trade** (Belarus alone was 20% pre-sanctions). Russia + Belarus + Canada = ~75% global potash.
- **Morocco**: 70% phosphate reserves; #1 source of P-fertilizer for India, Brazil, EU; OCP $14 bn 2025–27 capex plan. USAPP
- **Brazil/Argentina**: top soybean exporters (Brazil ~60% global soy export); Argentina = #1 soy meal/oil exporter.
- **Cote d'Ivoire/Ghana**: ~60% global cocoa; Cote d'Ivoire ~30% alone. World Population Review Indonesia #6. Lpem
- **Vietnam**: #2 rice exporter (4.72 mt H1 2025), Vietnam Law Magazine #1 robusta coffee, **99.6% of Thailand's pepper imports**. Nation Thailand VietnamPlus Robusta + black pepper near-global price-setter.
- **Thailand**: rice (declining; 3.73 mt H1 2025), Vietnam Law Magazine natural rubber #1.
- **Indonesia**: ~60%+ of global palm oil (with Malaysia ~85% combined); coffee (#3 robusta); cocoa.
- **Malaysia**: palm oil (~25–30%); rubber.
- **India**: 30% global rice exports (largest), 20% global generic drug exports, ISPE key APIs.
- **Madagascar**: ~75–80% global vanilla.

- **Specialty**: Iran (saffron 90%+), Sri Lanka (Ceylon tea, cinnamon), Philippines (coconut oil, abaca fiber).

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5. MANUFACTURED GOODS WITH CONCENTRATION RISK

- **Taiwan – TSMC**: ~70% of global foundry market Wccftech (some metrics 71%); **~90% of leading-edge nodes (3nm/5nm)**; MBI Deep Dives sole producer of <5nm logic for Apple, Nvidia, AMD, Qualcomm, Broadcom. Single-fab risk for some node-specific products. The most concentrated single-firm dependency in the global economy.
- **Netherlands – ASML**: 100% monopoly on EUV lithography (~$180m/Low-NA, ~$380m/High-NA per machine). Gotrade Customers: TSMC ~40% of EUV, Samsung, Intel. Gotrade Veldhoven is the single most strategically important industrial site in the world. China ~45% of 2024 ASML revenue Yahoo Finance (mostly DUV).
- **South Korea**:
  - **DRAM**: Samsung 34% + SK Hynix 36% (Q1 2025) Industryresearch = ~70% global; Korea total 70.5% Invest Korea Blackridge Research
  - **NAND**: Samsung 35% + SK Hynix-Solidigm 22% = 55–60% global Blocks and Files
  - **HBM**: SK Hynix ~70% of HBM market (Nvidia primary supplier). Blocks and Files
  - OLED panels (>90% global flexible OLED), Korean batteries (LG ES, Samsung SDI, SK On).
- **Japan**: ~90% photoresists and fluorinated polyimides, ~93% high-purity HF; semiconductor wafer (Shin-Etsu, SUMCO ~55% combined of 300mm wafers); machine tools (Fanuc, Yaskawa robots), aerospace titanium (Osaka Titanium, Toho, ATI = combined 2x VSMPO sponge capacity); carbon fiber Toray ~30%.
- **Germany**: machine tools (DMG Mori, Trumpf), automotive Tier-1s (Bosch, Continental, ZF), specialty chemicals (BASF, Covestro, Evonik), Siemens industrial automation.
- **Switzerland**: pharma (Roche, Novartis, Lonza), watches, precision (ABB, Mettler Toledo), Swatch movements (ETA monopoly).
- **Belgium**: Umicore (catalysts/recycling, 2nd-largest auto-catalyst maker), Bekaert (steel cord ~25–30%).

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6. GEOGRAPHICAL CHOKEPOINTS

Maritime

- **Strait of Hormuz**: 20 mb/d crude (≈20% global liquids consumption); U.S. Energy Information Administrat 1/5 of global LNG; Statista 30% of global helium; ~⅓ of global urea, 20–30% of ammonia exports. Wikipedia 84% Asia-bound. Wikipedia Closed/contested in 2026 Iran war. Bypass: Saudi East-West pipeline, UAE Fujairah pipeline ≈ 2.6 mb/d combined spare. U.S. Energy Information Administrat
- **Strait of Malacca**: 23.7 mb/d oil; Statista ~25–30% of global maritime trade by volume; ~20% by value. PubMed Central Single biggest aggregate-trade chokepoint. Lombok/Makassar/ The Strauss Center Sunda are partial alternates.
- **Bab el-Mandeb / Red Sea**: 4.2 mb/d oil; Visual Capitalist 12% global maritime trade pre-2024. Houthi attacks Nov 2023 → Sept 2025 (99 attacks). MarineLink Container traffic −86% in Q4 2025 vs 2023; bulkers −55%; crude tankers −32%; product tankers −19%. MarineLink Cape rerouting adds 10–14 days, 4,000 mi, ~$1m fuel per voyage. Wikipedia November 2025 ceasefire holding but conditional.
- **Suez Canal**: ~12% pre-crisis world trade → 9% in 2025; Coface container transits −90% YoY at peak; Coface Egypt revenue −22% in 2023. Isdo
- **Bosphorus / Turkish Straits**: ~3 mb/d oil The Strauss Center + ~10% global grain; ~50,000 vessels/yr; AA.com.tr only Black Sea exit.
- **Panama Canal**: 2.5–3% of global trade; GEP FY24 transits −29% (12,638→9,936) due to drought; Seatrade Maritime LNG transits crashed from ~26/month to ~4/month in H1 2025; Kuehne-nagel daily slots cut from 36 to 24 at trough (Feb 2024). GEP Now recovered but LNG hasn't returned.
- **Danish Straits**: 4.9 mb/d crude/products H1 2025 (~6% of global); U.S. Energy Information Administrat historically Russia-Europe.
- **Strait of Gibraltar**: gateway for ~15% of trade value into/out of Mediterranean.
- **Taiwan Strait**: ~20% global maritime trade by value; AI-chip flow choke; Luzon Strait similarly critical for cables.
- **South China Sea**: ~⅓ global shipping by value transits.
- **Cape of Good Hope**: not a chokepoint per se, but the alternative absorbing Red Sea/Suez bypass; +15 days Asia-Europe. U.S. Energy Information Administrat
- **Strait of Dover / English Channel**: ~15% of global trade by value transits north-south.
- **Strait of Otranto**: Adriatic gateway.

Land

- **Polish-Belarusian border (Małaszewicze)**: ~90% of China-EU rail freight transits this single rail terminal — Belarus closure or sanctions risk.
- **Khorgos (China-Kazakhstan)**: critical Belt-and-Road dry port.
- **Druzhba pipeline**: Russian crude to Hungary, Slovakia, Czechia (still active under exemptions).
- **Nord Stream 1/2**: destroyed Sept 2022.
- **TurkStream / Blue Stream**: Russian gas to Turkey/SE Europe.
- **BTC (Baku-Tbilisi-Ceyhan)**: 1 mb/d Caspian crude bypass of Russia/Hormuz.
- **Power of Siberia 1/2**: Russian gas to China; PoS-2 not yet built.
- **Yamal-Europe pipeline**: largely idle.
- **Trans-Caspian rail/ferry**: emerging Russia-bypass for Kazakh uranium and Caspian goods.
- **Alpine passes (Brenner, Gotthard, Mont Blanc)**: critical for European trucking; rail tunnels are single-points-of-failure.

Undersea cables (a 2024–2026 emergent chokepoint)

- **Red Sea / Bab el-Mandeb**: 15 major cables, ~25% of global Asia-Europe data. Gulf News Houthi anchor cuts (Feb 2024) damaged AAE-1, SEACOM/TGN-EA, EIG, IMEWE ORF Online — multi-month outages affecting India, Pakistan, Gulf, ~100m users in West/North Africa.
- **Strait of Hormuz**: 7+ major cables (Falcon, AAE-1, TGN-Gulf, SEA-ME-WE) — IRGC-linked Tasnim explicitly threatened these in 2026.
- **Luzon Strait**: between Taiwan and Philippines — China-US data corridor; Taiwan-strait scenario directly threatens.
- **Egypt land crossing**: Tagare estimates Egypt failure could take down 1/3 of global internet.
- **2Africa Pearls / SEA-ME-WE 6 Gulf Extension**: Meta paused Persian Gulf work in 2026; Submarine Networks Saudi terrestrial bypass via stc backbone now central.

Air freight hubs

- Memphis (FedEx), Hong Kong (HKIA — #1 cargo), Anchorage (transpacific transit), Doha, Dubai, Frankfurt, Shanghai PVG.

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7. VULNERABLE LOGISTICS NODES

- **Container ports (2024 TEU)**: Shanghai 51.5m (#1, first ever past 50m), Visual Capitalist Singapore 41.1m, Ningbo-Zhoushan 39.3m, Shenzhen 33.4m, Qingdao 30.9m, Guangzhou 26.1m, Container News Busan, Tianjin, Jebel Ali ~15.5m, Port Klang, **Rotterdam 13.8m,** American Journal of Transportati **Antwerp-Bruges 13.5m, Hamburg 7.8m**, American Journal of Transportation LA 10.3m / Long Beach 9.5m. China = 8 of top 20; ~55.6% of top-20 traffic. Upply **Hong Kong dropping** (13.7m, down from 16.6m in 2022) — alliance restructuring bypassing for Shenzhen/Guangzhou.
- **Single-source factories**:
  - ASML's Veldhoven HQ + Wilton MA mirror pod facility — single-source for EUV.
  - TSMC Hsinchu/Tainan Fabs 12/14/15/18 — only leading-edge silicon.
  - SK Hynix Icheon M16 + Samsung Pyeongtaek P3/P4 — HBM.
  - Stella Chemifa Osaka — semiconductor HF.
  - JSR Yokkaichi — EUV photoresist.
  - CBMM Araxá — niobium.
  - Eti Maden Bigadiç/Kırka/Emet — boron.
  - OCP Khouribga + Jorf Lasfar — phosphate.
  - QatarEnergy Ras Laffan — helium + 20% of global LNG.
  - VSMPO-Avisma Verkhnyaya Salda — aerospace titanium. RuAviation
  - Norilsk Nickel Polar Division — palladium and Russian nickel.
- **Refining clusters**: Houston Ship Channel (~5 mb/d), ARA (Antwerp-Rotterdam-Amsterdam ~3 mb/d), Singapore Jurong, Saudi Jubail/Ras Tanura, UAE Ruwais, Indian Jamnagar (Reliance, world's largest single refinery 1.24 mb/d).
- **Container shipping**: Top 3 alliance + MSC = ~80% of east-west capacity; MSC alone now ~20% world fleet.
- **Tanker fleet**: Russian "shadow fleet" 600+ aging tankers crucial for sanctioned crude flows; insurance/IACS concentration in P&I clubs.
- **Dry bulk**: iron-ore Capesize fleet concentrated, China-Australia route ~30% of Capesize tonne-miles.

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8. SCENARIO-BASED DISRUPTION MAPPING

Iran / Strait of Hormuz closure (live, 2026)

- **Hits**: 20% global oil, 20% LNG, 30% helium, ~⅓ urea, 20% ammonia, semiconductor noble gases bottleneck, undersea cables. Korean/Taiwanese chip risk first. Aluminum (Gulf = 9% of smelting + 18% of exports). Sulfur seaborne ≈⅓ global. **Cascade**: → Chinese sulfuric acid export ban → Chile/DRC copper, Indonesia HPAL, US/EU phosphate fertilizer.
- **Beneficiaries**: US shale crude (~13.7 mb/d capacity), US/Australia/Qatar (post-restart) LNG; ExxonMobil/North American Helium; Norwegian gas; Saudi East-West pipeline operators; Cape of Good Hope routing (SA ports); India refinery throughput.

Taiwan invasion/blockade

- **Hits**: ~90% leading-edge logic chips offline; HBM (SK Hynix in Korea, exposed to Korea-peninsula spillover); ASML demand shock; Apple/Nvidia/AMD/Qualcomm output crippled; auto chips; ALL container traffic via Taiwan Strait + South China Sea reroutes.
- **Beneficiaries**: Samsung (memory share gains), Intel Foundry, Japanese materials; SMIC/CXMT in China (if Beijing wins); GlobalFoundries (mature nodes); Texas Instruments analog.

Russia-NATO escalation

- **Hits**: remaining Russian gas to EU; titanium (Boeing/Airbus); palladium spike; Druzhba/TurkStream/Yamal flows; uranium enrichment (40% global SWU); Black Sea grain corridor; potash (Belarus). Baltic undersea cable risk.
- **Beneficiaries**: Norwegian Equinor; Qatar/US LNG; Cameco/Orano enrichment; Saskatchewan potash (Nutrien, Mosaic); Brazil/Australia ag.

Korean peninsula

- **Hits**: 70% global DRAM, 55% NAND; HBM chokepoint for AI; OLED displays; LG/Samsung/SK batteries; Busan port (5th largest); Ulsan refining cluster.
- **Beneficiaries**: Micron (Idaho, Hiroshima); Japanese semis; Samsung Texas; TSMC if logic-only.

Red Sea / Houthi attacks (ongoing, suspended Nov 2025)

- **Hits**: Asia-Europe container 25–35% cost premium sustained; DocShipper auto/electronics inventory; Egyptian Suez revenue; West/North Africa internet; LNG to

Europe.

- **Beneficiaries**: South African ports (Cape rerouting), Maersk/MSC if they hold capacity (lower so far); Egyptian land bridge; UAE Etisalat (cable traffic rerouting); air freight (DHL/FedEx).

Panama Canal water crisis

- **Hits**: US East Coast goods imports from Asia; LPG/LNG arbitrage US Gulf-to-Asia; West Coast intermodal pressure; agricultural exports (US grain).
- **Beneficiaries**: West Coast US ports (LA, Long Beach), BNSF/UP intermodal; rail-bridge providers (Maersk Panama land bridge); Suez (in normal times); Cape route.

China-India border / Himalayan

- **Hits**: limited direct trade; pharma API supply (India 65–70% APIs from China); Actizaindustry rare earth/magnet flow to India.
- **Beneficiaries**: Indian PLI scheme API plants; Vietnam/Thailand intermediates; US/EU pharma manufacturers.

Israel regional war expansion

- **Hits**: Bromine — South Korea 90–97.5% from Israel J2 Sourcing (HBr in DRAM/NAND etching). Mediterranean/Levant gas (Tamar, Leviathan). Suez approach. Jordan potash (Arab Potash, ~5% global).
- **Beneficiaries**: ICL (Israeli operations safe vs other producers), Albemarle bromine (Arkansas); Saudi/UAE diplomatic capital.

Sahel / West Africa coup belt

- **Hits**: Niger uranium (Orano); Mali gold; Burkina Faso gold; Cote d'Ivoire/Ghana cocoa risk if extended; West African oil (Nigeria production down).
- **Beneficiaries**: Kazakh uranium (Cameco/Kazatomprom); Canadian gold producers; Indonesian/Brazilian cocoa.

South China Sea incident

- **Hits**: ASEAN-China-Japan-Korea trade; Philippines/Vietnam supply; Strait of Malacca insurance.
- **Beneficiaries**: Lombok/Makassar transit; India ports (Mumbai, Chennai); Cape route fallback.

Arctic competition

- **Hits**: Northern Sea Route disputes; Russian Arctic LNG-2 (Yamal); undersea cable Arctic alternatives (Far North Fiber).
- **Beneficiaries**: Norwegian/Arctic operators; Greenland REE (Tanbreez, Kvanefjeld).

African coup belt instability

- **Hits**: see Sahel + DRC mining (Goma 2024 conflict ~1,400km from Copperbelt — limited direct cobalt impact but precedent risk); Cobalt Institute Ethiopia rail.
- **Beneficiaries**: South African mining, Botswana, Morocco's African phosphate diplomacy.

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9. HIDDEN DEPENDENCIES AND CASCADE EFFECTS

1. **Iran sulfur → China sulfuric acid → Chilean copper (SX-EW)**: 20% of Chile copper output requires acid; 1 mt+/yr from China MINING.COM — already triggered (May 2026 ban). HIDDEN DEPENDENCY: also hits DRC/Zambia copper-cobalt and Indonesian HPAL nickel — i.e., a Middle East war directly tightens EV battery raw materials.
2. **Hormuz closure → Qatar helium → MRI uptime + Korean chip yields**: 30% of helium gone; hospitals 2–4 weeks of inventory; Samsung/SK 6 months. ~200 cryogenic containers stranded.
3. **Myanmar civil war (Kachin) → Chinese heavy REE → global EV traction motors**: Kachin Independence Army control of Panwa region threatens the 70% feedstock China imports. Investing News Network Drives dysprosium/terbium volatility.
4. **Ukrainian neon (50–80% global) → ASML lithography → TSMC yields**: 2014 7–10x; 2022 partial offset by stockpiles.
5. **Russia titanium (VSMPO) → Boeing/Airbus airframes**: Pre-2022 80% of Boeing, 60% of Airbus titanium. Quest Metals Sponge cut from 32 kt to 17 kt; Japan (Osaka/Toho/ATTM ~2x VSMPO sponge capacity) is the prime beneficiary.
6. **Israeli bromine → Korean DRAM etching**: 97.5% of Korean bromine imports from Israel — most ME-dependent input in Korea's "14 critical materials" list. All-Weather Media
7. **Morocco OCP → African food security + EU/India fertilizer**: Single-country single-firm 70% reserve concentration. War or maritime disruption around Casablanca/Jorf Lasfar = global P shortage.
8. **South Africa rhodium → ICE catalytic converters → automotive output**: Rhodium #2 USGS critical mineral 2025, ~80% from a few SA operations. Eskom power cuts already

structurally constrain.

9. **Madagascar vanilla**: 75–80% global → global ice cream/flavor/fragrance industry hostage to single-island weather and political risk.
10. **Indonesian palm oil → biodiesel mandates + global cooking oil**: Indonesia + Malaysia ~85%; 2022 export ban triggered global edible-oil price spike.
11. **Single Niobium producer (CBMM Araxá)**: 77.4% global market share; high-strength low-alloy steels (auto, oil&gas pipelines, aerospace) and emerging XNO niobium-titanate battery anodes Global Market Insights — single-fire risk.
12. **Eti Maden Turkey boron → fiberglass + permanent magnets**: 60–73% global; included in NdFeB magnet; agricultural micronutrient. State-controlled, geopolitically exposed.
13. **Solar polysilicon Xinjiang → 40–45% global solar-grade poly + 95% modules use Chinese wafers (97% global wafer)**: any UFLPA escalation or Xinjiang sanction = solar deployment slowdown globally.
14. **Stella Chemifa (Osaka) high-purity HF**: ~63% global semiconductor-grade HF; Globalmarketmonitor single-firm earthquake/fire risk equal to many country risks.
15. **Belarus + Russia potash → 40% of global**: Indian rice yields, Brazilian soy yields are exposed.
16. **Indian generic APIs → Western healthcare**: 40% of US generic drugs, 25% of UK medicines from India; Drug Patent Watch but India imports 65–70% of its KSMs/intermediates from China Actizaindustry — China-India border or pharma export restriction = global drug shortages.
17. **Bekaert Belgium tire steel cord**: ~25–30% global; tire industry vulnerability.
18. **Single-source aerospace fasteners (Lisi/Howmet/PCC)**: Boeing's recurring delivery delays trace partly to fastener shortfalls.
19. **Copper smelting + sulfuric acid co-production**: smelters produce sulfuric acid as byproduct Bloomberg — copper concentrate trade controls fertilizer chain in non-obvious ways.
20. **ASML Wilton MA optics for EUV mirrors via Carl Zeiss SMT (Oberkochen, Germany)**: Zeiss's single Oberkochen facility is THE optical bottleneck for EUV, even further upstream than ASML.

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10. INVESTMENT IMPLICATIONS — BOTTLENECK

BENEFICIARIES

Dependency	Primary Risk	Alternative Producer	Time-to-Replace	Strategic Stockpile?
Heavy REE (Dy/Tb)	China 99%	Lynas (Mt Weld), MP Materials, REalloys, Iluka, Northern Minerals (Browns Range Q4 2025 FS), Investing News Network Madagascar projects	5–10 yr	US DLA holds limited; China stockpile large
Light REE (Nd/Pr)	China ~60%	Australia 15–20%, Brazil potential	3–5 yr	Yes
Niobium	CBMM Brazil 77%	Magris Niobec (Canada 8%), NioCorp (US dev)	5–7 yr	US DLA strategic
Cobalt	DRC 72%	Indonesia rising to 22% by 2030; Yahoo Finance Australia/Canada 8% by 2035; LFP substitution underway	2–5 yr	China SRB stockpiling
Lithium	Chile 23% / Australia 50% mined	Argentina (21% CAGR); Bolivia; Africa	3–5 yr greenfield	Limited
Nickel (battery)	Indonesia 50%	Philippines, New Caledonia, Australia; HPAL substitution risks	3–5 yr	Limited
Palladium	Russia 40%	South Africa; Pt-for-Pd substitution accelerated; Phoenix Refining recycling	2–4 yr	US DLA, JM, Heraeus
Platinum/Rhodium	South Africa 71%/80%	Russia, Zimbabwe rising; thrifting	5+ yr	Limited
Titanium aerospace	VSMPO ~30%	Japan (Osaka, Toho, ATTM combined 2x VSMPO), US ATI/Howmet	2–4 yr; partly done	US DPA Title III

Niobium sub-grade	none viable	Vanadium partial sub in some applications	n/a	n/a
Helium	US 43% / Qatar 30% / Algeria 7%	ExxonMobil (LaBarge), North American Helium, Helix, Blue Star, Air Liquide German storage; Russia (sanctioned)	3–5 yr new; cannot stockpile cheaply	US Federal Reserve fully privatized 2024 — buffer is gone
Neon	Ukraine 50–70%	Korea/China ramping; recycle	1–3 yr	Corporate
Photoresists / fluorinated PI	Japan 90%	None at scale; Korean local catch-up	5+ yr	Corporate
High-purity HF	Japan 93%	Foosung (Korea), Chinese DFD scaling	3–5 yr	Corporate
Phosphate	Morocco 70% reserves	Russia (sanctioned), China, US (Mosaic), Tunisia	Decades to replace reserves	India strategic, US
Potash	Canada/Russia/Belarus	Canada (Nutrien) flexible upside, Jordan, Laos, Israel (ICL)	3–7 yr greenfield	Limited
Boron	Turkey 73% reserves / 60% supply	Rio Tinto US Borax (Boron, CA ~30%), Argentina (Quiborax), Chile	Decades	Limited
Sulfuric acid	China 40%	None at scale; copper smelter byproduct chain	2–5 yr	n/a
Tungsten	China 80%	Almonty Sangdong (Korea, restart 2025), Vietnam (Masan Nui Phao), Portugal	3–5 yr	US DLA limited
Antimony	China 48% mining +	Bolivia, Tajikistan, Perpetua (US Stibnite	3–7 yr	Almost

	refining	Mine, 2027+)		none in US
Gallium / Germanium	China ~99% / 60%	Recycle from Zn/Cu refining; Korea/Japan/Russia	3–5 yr	Limited
Graphite battery	China 70% mine, 90% spherical	Syrah Vidalia (LA), Northern Graphite, NextSource Madagascar, Nouveau Monde Quebec	3–5 yr	None at scale
Magnesium	China 95%	Israel (Dead Sea), Brazil, Russia, US restart projects	5+ yr	n/a
EUV lithography	ASML 100%	None	10+ yr if ever	n/a
Leading-edge logic	TSMC 90%	Samsung Foundry, Intel Foundry 18A (2025–26 ramp); TSMC Arizona	5–10 yr	n/a
HBM	SK Hynix 70%	Samsung, Micron	1–3 yr	Corporate
Aerospace titanium	VSMPO ~30%	Japan, US (above)	Partly done	US DPA
Aramid/carbon fiber	Top 5 = 37–75%	DowAksa, Hyosung, China	3–5 yr	n/a
Cocoa	Cote d'Ivoire 30%	Indonesia, Ghana, Ecuador	5+ yr (tree maturity)	Industry
Vanilla	Madagascar 75%	Indonesia, Uganda, synthetic vanillin	3–5 yr	Industry
Coffee robusta	Vietnam #1	Brazil, Indonesia, Uganda	3–5 yr	ICE warehouse
Pepper	Vietnam ~99% of Thai imports	India, Brazil, Indonesia	2–3 yr	Limited

Macro positioning themes

1. **"Second-best producer" trade**: every Chinese export control is a forced re-rating event for the #2 producer outside China. Almonty (W), Perpetua (Sb), Lynas/MP Materials (REE), Syrah/Nouveau Monde (graphite), Iluka (zircon/REE), NioCorp (niobium).
2. **Sulfuric acid / fertilizer cascade**: long Mosaic, Nutrien, OCI, ICL, Yara, OCP IPO (when it comes); long phosphate rock juniors; expect Chilean copper margin compression.
3. **Helium pure-plays**: ExxonMobil (LaBarge dominant), Air Products (mid-stream distribution), Air Liquide (storage in Germany), North American Helium (private), Helix Exploration, Blue Star Helium, Avanti Energy, Pulsar Helium.
4. **Aerospace titanium re-shoring**: ATI Inc (US), Howmet, Osaka Titanium, Toho Titanium, Allegheny.
5. **Japanese specialty chip materials as China-substitution kings**: JSR (taken private by JIC 2023), Tokyo Ohka Kogyo, Shin-Etsu Chemical, Sumitomo Chemical, Stella Chemifa.
6. **Korea memory cycle + HBM dominance**: SK Hynix (HBM 70%), Samsung; Micron as the China-tariff-protected US play.
7. **TSMC + ASML duopoly**: the "national security" pricing power thesis — both are structurally pricing more like utilities-with-monopoly-rents than cyclicals.
8. **Niobium**: CBMM (private), NioCorp (NB; US, DPA-funded), Magris (private). Watch CMOC's nascent niobium rivalry.
9. **OCP IPO when listed; phosphate rock juniors (PHX, Itafos, Arianne)**.
10. **Eti Maden is state-owned (no pure listing)** but Rio Tinto US Borax exposure via Rio Tinto.
11. **Maritime risk premium**: war-risk insurers (Lloyd's syndicates), tanker pure-plays (Frontline, Scorpio, Euronav) for tonne-mile expansion when chokepoints close.
12. **Cape of Good Hope beneficiaries**: South African ports (Transnet listed peers), Mauritius-based SBM Offshore.
13. **Strategic stockpile situations**:

- **US Defense Logistics Agency (DLA Strategic Materials)**: maintains stockpiles of antimony, beryllium, germanium, manganese, niobium, tantalum, tungsten — but most are below thresholds; antimony is critically low.
- **China's State Reserve Bureau**: large stockpiles of cobalt (4% of 2023 demand purchased in restart of strategic buying), Cobalt Institute copper, aluminum, REE, grains.
- **US Strategic Petroleum Reserve**: ~370 mb (down from 700 mb peak).

- **Helium**: US Federal Helium Reserve fully privatized 2024 — no remaining public buffer.

14. **The "obscure derivative" of every shock**: in a sulfuric-acid squeeze, copper smelters with captive acid markets benefit; in a helium squeeze, helium-recovery technology vendors (Quantum Design, Linde recovery systems) re-rate; in a Hormuz closure, alternative pipeline operators (Saudi Aramco East-West, BTC) run at capacity premium.

CAVEATS

- **Source quality is uneven**: market percentages vary by source and year. Authoritative anchors are USGS Mineral Commodity Summaries 2025, IEA, EIA, OPEC ASB 2025, IGU LNG Report 2025, Cobalt Institute, World Nuclear Association. Several percentages quoted (e.g., "70%/80% Boeing/Airbus titanium dependency on VSMPO" pre-2022) are widely cited but originate from a small number of trade-press estimates.
- **2026 Iran-war narrative**: the Bloomberg/SCMP/Akin Gump/Foreign Policy/Al Jazeera/Scientific American sources describe an active or recent disruption (Strait of Hormuz contested, Ras Laffan damaged, Chinese sulfuric acid export halt effective May 2026, helium force majeure, tanker rerouting). I have presented these as reported events; readers should verify the current status of the ceasefire, Hormuz transit recovery, and Qatar production restart, which were in flux through April 2026 and may have evolved further.
- **Speculative vs realized**: critical-minerals price moves are real (antimony +2,600%, dysprosium ex-China premiums 2.5–3x). Forward projections (Indonesia overtaking DRC cobalt by 2040; 80% China graphite share through 2035) are consensus forecasts, not facts.
- **Chinese export-control regime is volatile**: the November 2025 US-China deal paused the October 9 expanded REE controls for one year (through Nov 10, 2026), Pillsbury Winthrop Shaw Pittman but tungsten/antimony/silver licensing for 2026–27 remains in force. Bilateral status can re-tighten with little notice.
- **Single-firm/single-facility risks have asymmetric tails**: VSMPO sponge, Stella HF, Eti Maden, OCP, CBMM, ASML/Zeiss are all individually capable of a six-month-plus global supply outage from internal failure (fire, strike, sanction) — independent of the named geopolitical scenarios.
- **Substitution timelines are long**: heavy REE (5–10 yr), niobium (5–7 yr), high-purity HF (3–5 yr). Investors should size positions assuming the bottleneck holds even under aggressive substitution.
- **Several cited "obscure" findings (e.g., Bekaert tire cord, Madagascar vanilla,**

Vietnamese pepper to Thailand) are real but smaller in dollar terms than the headline minerals/energy risks; treat them as pair-trade or thematic overlays rather than core positions.

- **Country-level political risk is non-linear**: Guinea bauxite (military government), Niger uranium (post-coup), DRC cobalt (artisanal/conflict overlap), Kachin REE feedstock to China (active civil war) are all on tail-risk knife-edges and can move faster than commodity-price models capture.